Interactive MD simulation of a protein

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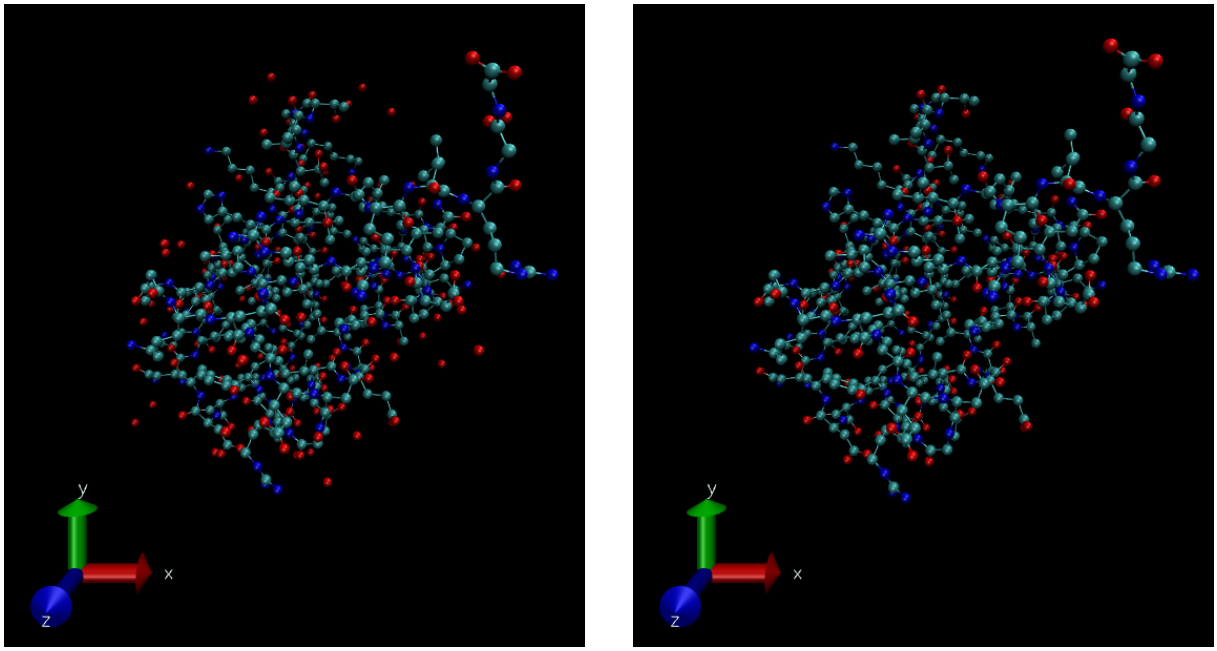
Question 1

What was the experimental technique employed to determine the protein structure? What is the resolution (in angstrom) of the structure? How many atoms has the protein?

If we consider the information provided by the Protein Data Bank, the experimental technique employed to determine the protein structure was X–ray diffraction. Consequently, as shown in Figure 1a, there are no hydrogen atoms in the stucture. In addition, the structure has a resolution of 1.80 angstroms and contains 660 atoms.



(a) Representation of UBQ1 in VMD.

(b) Representation of UBQ1 after the selection process

Figure 1

Question 2

What are the atoms that are not part of the protein? How do you save only the protein atoms?

In order to obtain a .pdb file containing only protein atoms, a tool in VMD is used to select them. This tool allows us to define the desired atomic selection, such as *protein*, *all not protein*, *nucleic acids*, etc.

By using the selection *protein*, a representation containing only protein atoms is obtained, as shown in Figure 1b. The total number of atoms decreases from 660 to 602 compared to the previous structure. The atoms removed during the selection process correspond to oxygen atoms

associated with the solvent (water), which were interacting with the protein through hydrogen bonds. It should be remembered that hydrogen atoms are not present, as the structure was determined by X-ray diffraction.

Question 3

What is the difference between both files?

The topology file provides information about topological structure of the molecule. Specifically, it contains the atoms and they are connected through covalent bonds. It also defines aminoacids and the parcial charge of each atom. In conclusion, the molecular structure is defined in this file.

On the other hand, the parameter file contains the necessary parameters to describe the molecules. It includes bond force constant, angles, dihedrals, etc.

Question 4

You have generated a new pdb file. Compare the new and the old pdb. What is the main difference?

The main difference betwen both files lies in the number of atoms. The total number of atoms increases significantly from 602 to 1231. This increase is due to the addition hydrogen atoms, which are now included in the structure. The new structure is shown in Figure 2. Looking at the details, many new atoms added can be observed in white, corresponding to hydrogen atoms.

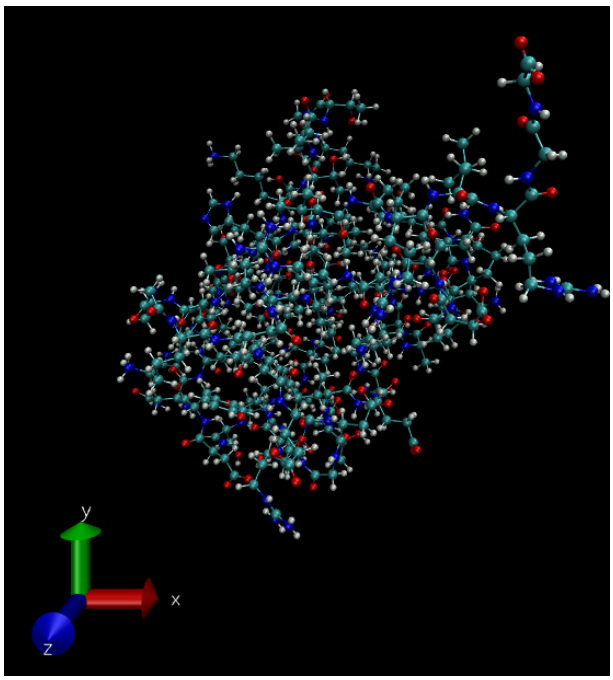


Figure 2: Representation of the *psf* file.

Question 5

Plot the energy as a function of time and briefly comment the results. Also do the same with the temperature.

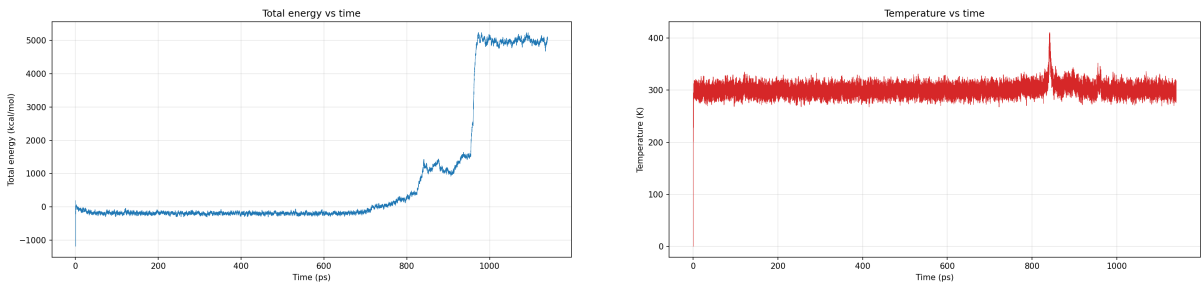
The Figure 3b provides information about the temperature evolution over the simulation time.

Overall, it is clear that the temperature remains relatively stable around 300 K, except for a peak at the beginning of the simulation. The sharp increase observed at time 0 is due to the initial stage of the simulation, where the system reaches thermal equilibrium.

In addition, around 840 ps, the temperature reaches a peak of 410 K, which is consistent with a corresponding peak in the energy vs. time plot (Figure 3a).

On the other hand, the energy vs. time plot is shown in Figure 3a. As can be seen, the energy remains stable until approximately 700 ps. After this point, it starts to increase steadily except for a peak at 840 ps. Before 1000 ps, the energy increases significantly from approximatelly 1800 to 5000.

The sharp increase in potential energy is associated with the external forces applied during the simulation. When a force is applied to an atom, work is performed on the system. As can be seen in both graphs, when an external force applied, the total energy increases and reaches a new higher value, whereas the temperature rturns to its previous value.



(a) Energy vs time

(b) Temperature vs time

Figure 3

Question 6

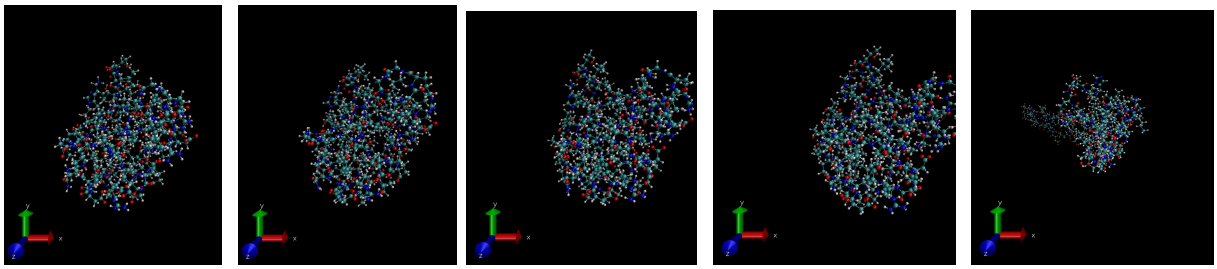
Show an image of the protein in your preferred representation before starting the simulation, before starting the application of the force and at the end of the simulation. Comment the images

The evolution of the molecule during the simulation, in terms of frames, is shown in Figures 4 and 5. Figure 4 covers the simulation from frame 500 to 4000, whereas Figure 5 covers frames from 4000 to 5000. The simulation consists of a total of frames.

Before the application of the external force, a contraction and expansion motion is observed in the protein. This type of motion is common in proteins, as they are known to be dynamic rather than static. According to Figure 3a, the external force is applied around 700 ps. This time value corresponds to approximately frame 3500, based on the simulation parameters(timestep and DCD frequency, equal to 2 and 100, respectively).

When the force is applied, the molecule begins to move along the direction of the applied force.

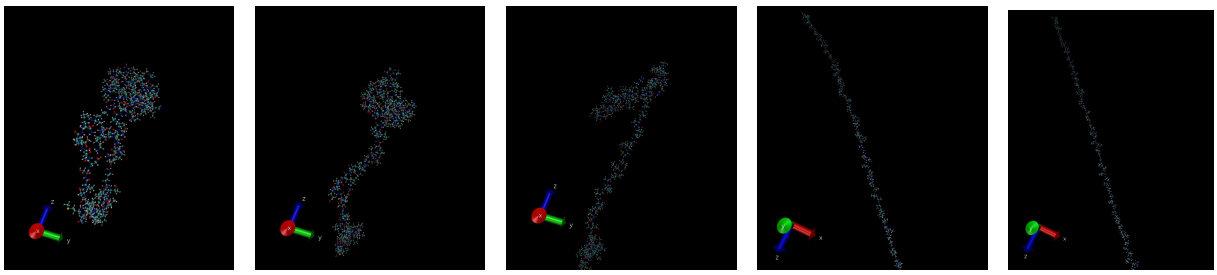
This displacement can be observed between the frames 3000 and 4000.



(a) 500 (b) 1000 (c) 1500 (d) 3000 (e) 4000

Figure 4

On the other hand, the elongation of the protein is illustrated in Figure 5, which covers the simulation frames from 4000 to 5000. In order to obtain this linear structure, several forces were applied to different atoms along different directions.



(a) 4000 (b) 4100 (c) 4300 (d) 4800 (e) 5000

Figure 5

Figure 6 illustrates a magnified view of the protein in the final step. As can be seen, the molecule is practically linear and amino acids are arranged in series.

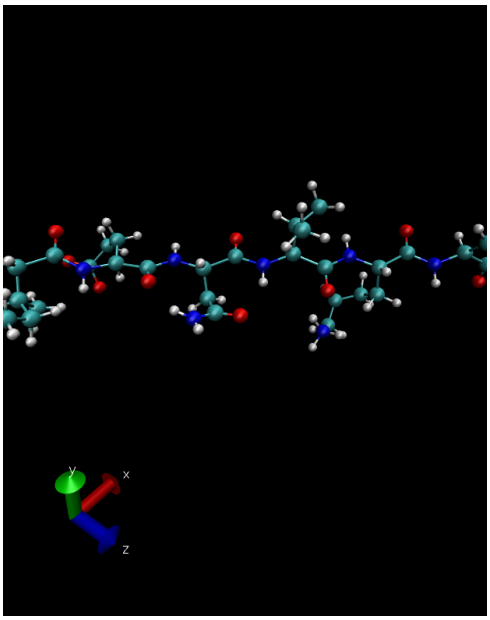


Figure 6: 5610